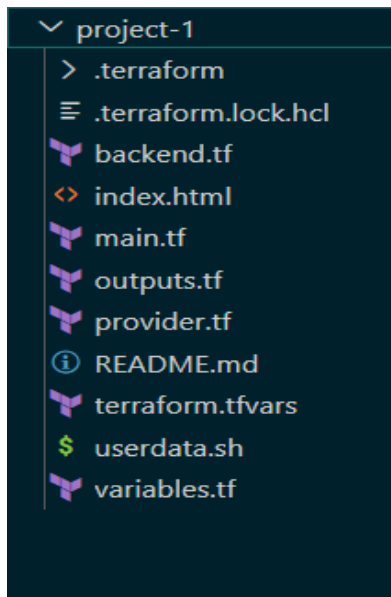
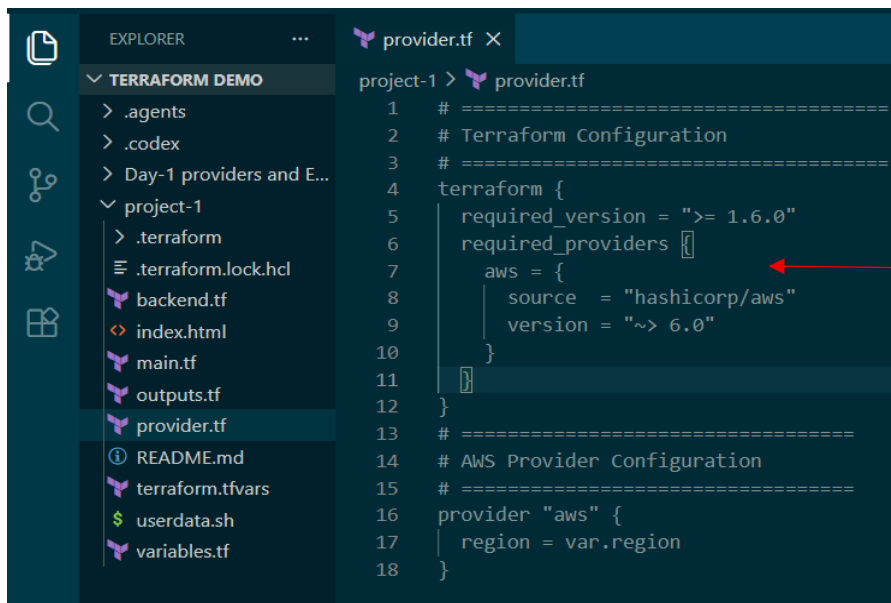


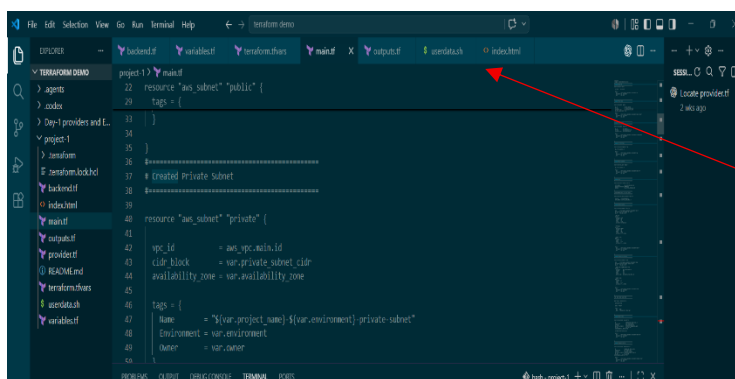
BASIC REPORT OF PROJECT -1



STEP-1 I created project-1 folder in that folder I created multiple files like main.tf, backend.tf, etc.



STEP-2 Configured the AWS provider and specified the required Terraform version and AWS provider version



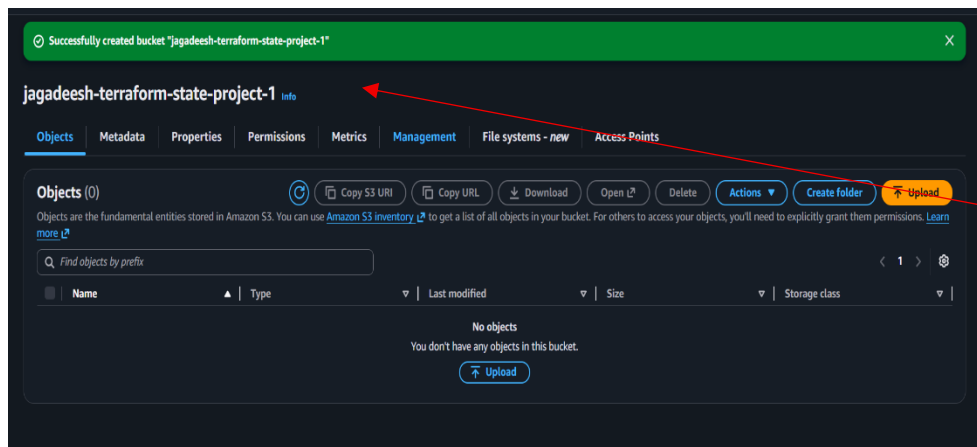
STEP-3 Configured the backend, variable terraform. tfvars, output and main.tf and userdata.sh

PROJECT-1

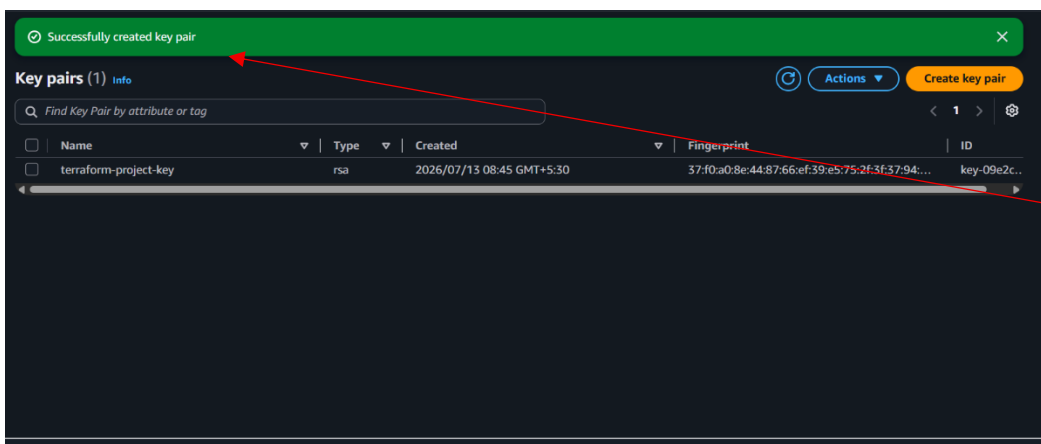
```
PS C:\Users\jagad\OneDrive\Desktop\DevOps\terraform demo\project-1> terraform version
Terraform v1.15.7
on windows_amd64

Your version of Terraform is out of date! The latest version
is 1.15.8. You can update by downloading from https://developer.hashicorp.com/terraform/install
PS C:\Users\jagad\OneDrive\Desktop\DevOps\terraform demo\project-1> |
```

STEP-4 Updated
Terraform
Version



STEP-5
created s3
bucket in AWS
console



STEP-6 created
key pair in EC2

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\jagad>curl ifconfig.me
103.208.231.57
C:\Users\jagad>|
```

STEP-7 for my
Ip address I
checked in cmd

PROJECT-1

```
PS C:\Users\jagad\OneDrive\Desktop\DevOps\terraform demo> Error: creating EC2 Instance: operation error EC2: RunInstances, https response error StatusCode: 400, RequestID: 0bb84d10-e16e-40af-bf22-63aad34e2744, api error InvalidParameterCombination: The specified instance type is not eligible for Free Tier. For a list of Free Tier instance types, run 'describe-instance-types' with the filter 'free-tier-eligible=true'.|| with aws_instance.web_server,| on main.tf line 203, in resource "aws_instance" "web_server":| 203: resource "aws_instance" "web_server" {| Error: creating EC2 Instance: operation error EC2: RunInstances, https response error StatusCode: 400, RequestID: 79ffdb8c-63c2-4ad9-8ddb-ab29b6991b85, api error InvalidParameterCombination: The specified instance type is not eligible for Free Tier. For a list of Free Tier instance types, run 'describe-instance-types' with the filter 'free-tier-eligible=true'.|| with aws_instance.database_server,| on main.tf line 225, in resource "aws_instance" "database_server":| 225: resource "aws_instance" "database_server" {
```

STEP-8 check instant type whether it is free tier or not I given instant type as t2. micro it is showed error because it not frees then again, I change to t3. micro

Instance type	Free ...	vCPUs	Architecture	Memory (GiB)	Storage (GB)	Storage type	Network performance
t2.micro	false	1	i386, x86_64	1	-	-	Low to Moderate
t3.micro	true	2	x86_64	1	-	-	Up to 5 Gigabit

1 VPC (e.g. 10.0.0.0/16)

NETWORK (VPC)

The screenshot shows the AWS VPC console. On the left, the 'VPC dashboard' sidebar is visible. The main area shows 'Your VPCs (1/2)' with a table listing VPCs. The selected VPC is 'terraform-project-dev-vpc' with VPC ID 'vpc-08ff8fc61f33f6034' and state 'Available'. Below this, the 'CIDRs' tab is selected, showing an 'IPv4 CIDR' of '10.0.0.0/16' with status 'Associated'.

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4
terraform-project-dev-vpc	vpc-08ff8fc61f33f6034	Available	-	-	Off	10.0.0.0/16

Address family	CIDR	Status
IPv4	10.0.0.0/16	Associated

1 Public Subnet (for the web server)

- 1 Private Subnet (for the database)

The screenshot shows the AWS VPC console with the 'Subnets' tab selected. It displays a table of subnets for the VPC 'terraform-project-dev-vpc'. There are two subnets: 'terraform-project-dev-public-subnet' (Public) and 'terraform-project-dev-private-subnet' (Private). Both are in 'Available' state.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
terraform-project-dev-public-subnet	subnet-0ced1f02dbe292ac7	Available	vpc-08ff8fc61f33f6034 terraf...	Off	10.0.1.0/24
terraform-project-dev-private-subnet	subnet-0e29bc1324e687077	Available	vpc-08ff8fc61f33f6034 terraf...	Off	10.0.2.0/24

Internet Gateway attached to the VPC

The screenshot shows the AWS Management Console for the 'Internet gateways' section. The left sidebar contains the 'VPC dashboard' with a 'Filter by VPC' dropdown. The main content area displays a table of Internet gateways. The first gateway, 'terraform-project-dev-igw', is highlighted. Below the table, the details for this gateway are shown, including its ID, state (Attached), VPC ID, and owner.

Name	Internet gateway ID	State	VPC ID	Owner
terraform-project-dev-igw	igw-027ab774ca4129331	Attached	vpc-08ff8fc61f33f6034 terraform-proj...	258233813982
-	igw-Dd9d76f22e7402fcb	Attached	vpc-0a801ac797b68c78b	258233813982

igw-027ab774ca4129331 / terraform-project-dev-igw

Details

Internet gateway ID igw-027ab774ca4129331	State Attached	VPC ID vpc-08ff8fc61f33f6034 terraform-project-dev-vpc	Owner 258233813982
--	-------------------	---	-----------------------

Route Table for the public subnet -> 0.0.0.0/0 via IGW

The screenshot shows the AWS Management Console for the 'Route tables' section. The left sidebar contains the 'VPC dashboard' with a 'Filter by VPC' dropdown. The main content area displays a table of Route tables. The third route table, 'terraform-project-dev-public-rt', is highlighted. Below the table, the routes for this route table are shown, including the destination, target, status, propagated status, and route origin.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0c88514daa8ff8616	-	-	Yes	vpc-0a801ac797b68c78b
-	rtb-0bc42845f51f545b7	-	-	Yes	vpc-08ff8fc61f33f6034 terraform-proj...
terraform-project-dev-public-rt	rtb-0fa41f10e35634888	subnet-0ced1f02dbe292...	-	No	vpc-08ff8fc61f33f6034 terraform-proj...

rtb-0fa41f10e35634888 / terraform-project-dev-public-rt

Routes (2)

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-027ab774ca4129331	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

PROJECT-1

Public subnet must auto-assign public IPs

The screenshot shows the AWS VPC console. The left sidebar has a 'VPC dashboard' section with a 'Subnets' link. The main content area shows a list of subnets. The first subnet, 'terraform-project-dev-public-subnet', is selected. Below the list, the details for this subnet are shown. The 'Auto-assign public IPv4 address' checkbox is checked, which is highlighted by a red arrow from the callout box above. Other details include the availability zone 'ap-south-1', network ACL 'acl-07df42272ee24787a', and VPC 'vpc-08ff8fc61f33f6034'.

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
terraform-project-dev-public-subnet	subnet-0ced1f02dbe292ac7	Available	vpc-08ff8fc61f33f6034 terraform-project-dev-vpc	Off	10.0.1.0/24
-	subnet-09eba719256f8b844	Available	vpc-0a801ac797b68c78b	Off	172.31.16.0
terraform-project-dev-private-subnet	subnet-0e29bc1324e687077	Available	vpc-08ff8fc61f33f6034 terraform-project-dev-vpc	Off	10.0.2.0/24
-	subnet-0f8aa51953831c741	Available	vpc-0a801ac797b68c78b	Off	172.31.0.0/24

subnet-0ced1f02dbe292ac7 / terraform-project-dev-public-subnet

- Availability Zone: ap-south-1
- Network ACL: acl-07df42272ee24787a
- Auto-assign customer-owned IPv4 address: No
- IPv6 CIDR reservations: No
- Network border group: ap-south-1
- Default subnet: No
- Customer-owned IPv4 pool: No
- IPv6-only: No
- VPC: vpc-08ff8fc61f33f6034 | terraform-project-dev-vpc
- Auto-assign public IPv4 address: Yes
- Outpost ID: -
- Hostname type: -
- Route table: rtb-0fa41f10e35634888 | terraform-project-dev-public-rt
- Auto-assign IPv6 address: No
- IPv4 CIDR reservations: -
- Resource name DNS A record: -

SECURITY

Web Security Group:
Allow SSH (22) -> only from YOUR Ip
Allow HTTP (80) -> from 0.0.0.0/0

The screenshot shows the AWS Security Groups console. The left sidebar has a 'Network & Security' section with a 'Security Groups' link. The main content area shows a list of security groups. The first security group, 'terraform-project-dev-web-sg', is selected. Below the list, the details for this security group are shown. The 'Inbound rules' section is expanded, showing two rules: one for HTTP (80) from 0.0.0.0/0 and one for SSH (22) from 103.208.231.57/32. The SSH rule is highlighted by a red arrow from the callout box above.

Name	Security group ID	Security group name	VPC ID
terraform-project-dev-web-sg	sg-0dd6be22f700f7ec4	terraform-project-dev-web-sg	vpc-08ff8fc61f33f6034
-	sg-0263d37a45fbx4816	default	vpc-08ff8fc61f33f6034
terraform-project-dev-db-sg	sg-08cd0a5e1c308bca0d	terraform-project-dev-db-sg	vpc-08ff8fc61f33f6034

sg-0dd6be22f700f7ec4 - terraform-project-dev-web-sg

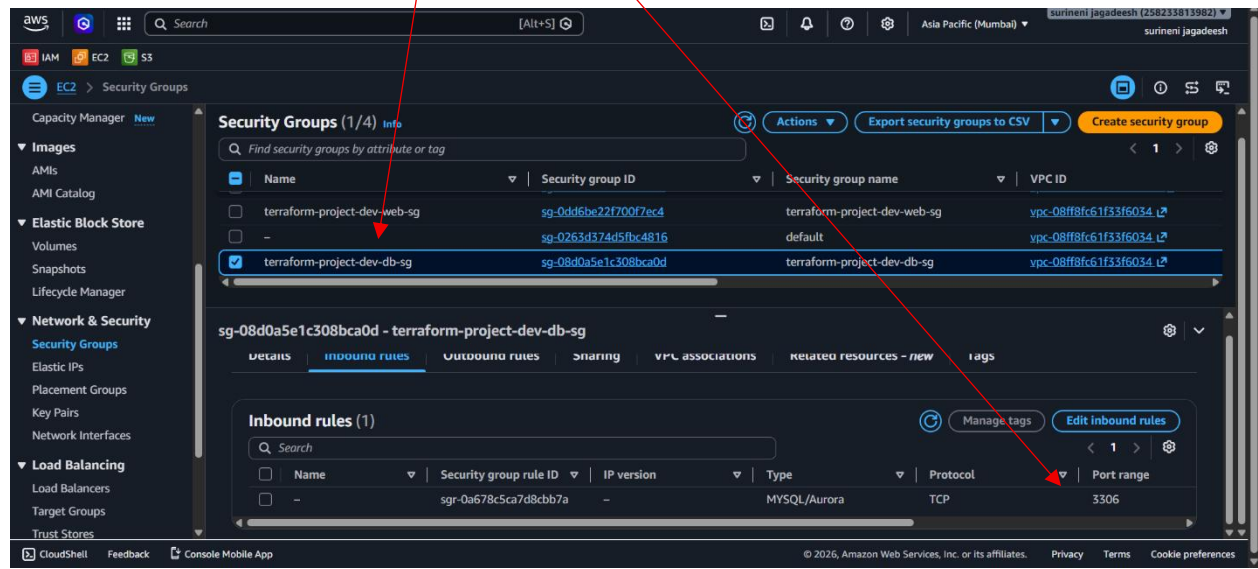
Inbound rules (2)

Security group rule ID	IP version	Type	Protocol	Port range	Source
sg-0a31f9eadbcd09e7d	IPv4	HTTP	TCP	80	0.0.0.0/0
sg-0fa7476ce93b3735a	IPv4	SSH	TCP	22	103.208.231.57/32

PROJECT-1

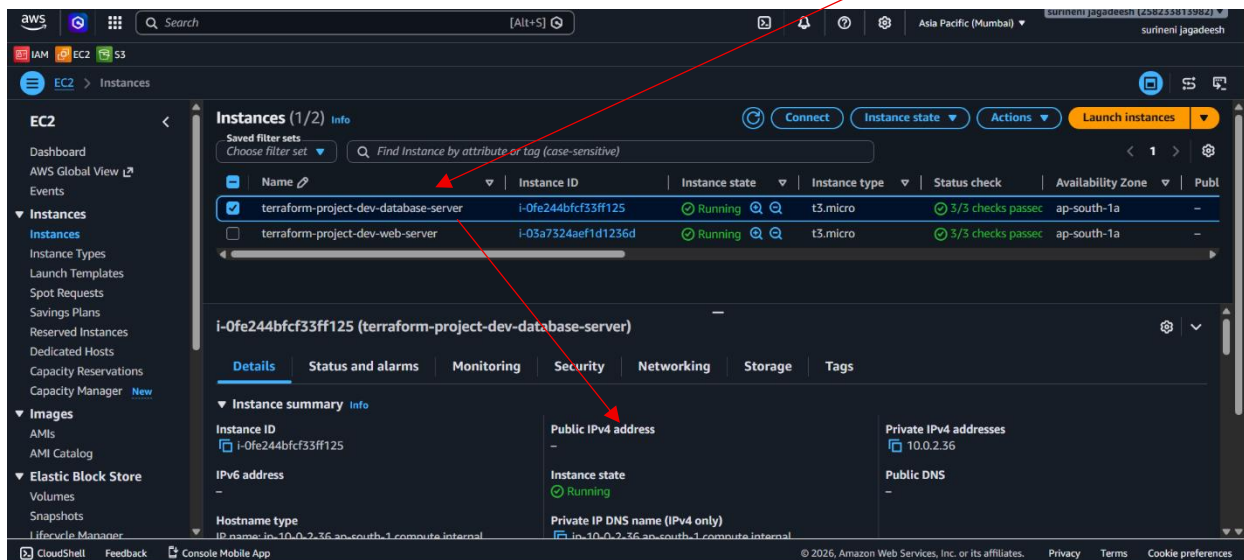
DB Security Group:

Allow MySQL (3306) -> ONLY from the Web Security Group



DB Security Group:

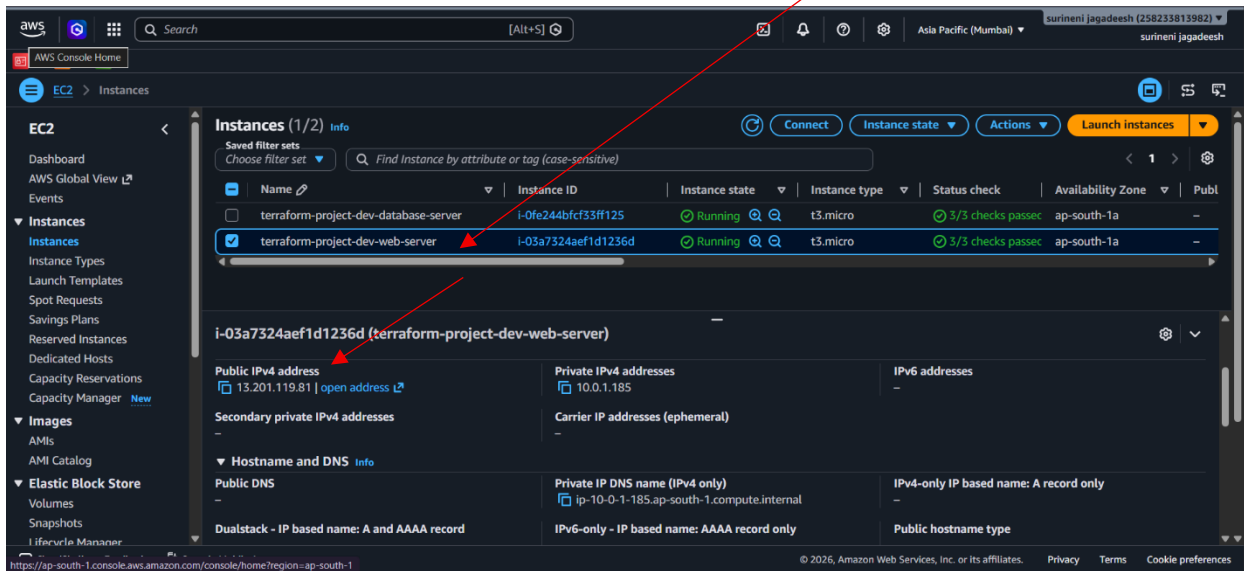
NO public access at all



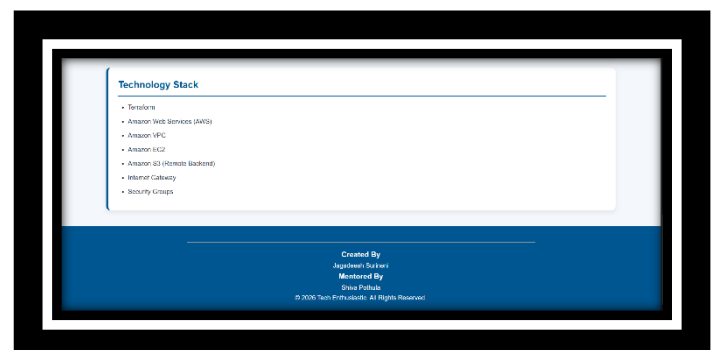
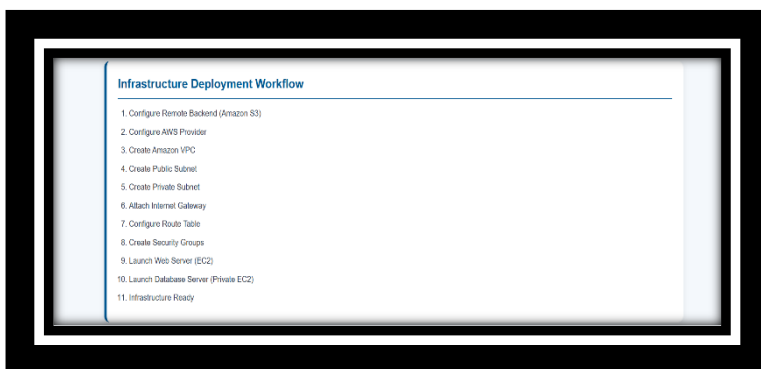
PROJECT-1

COMPUTE

1 EC2 instance in the PUBLIC subnet (the web server)

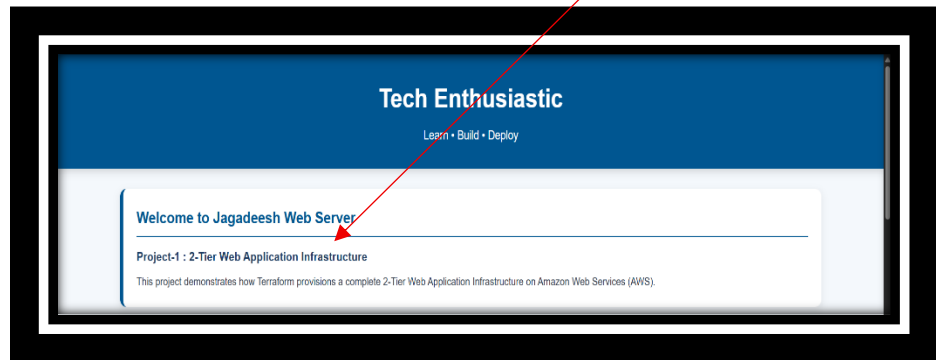


Install a web server using user data (Apache or Nginx)



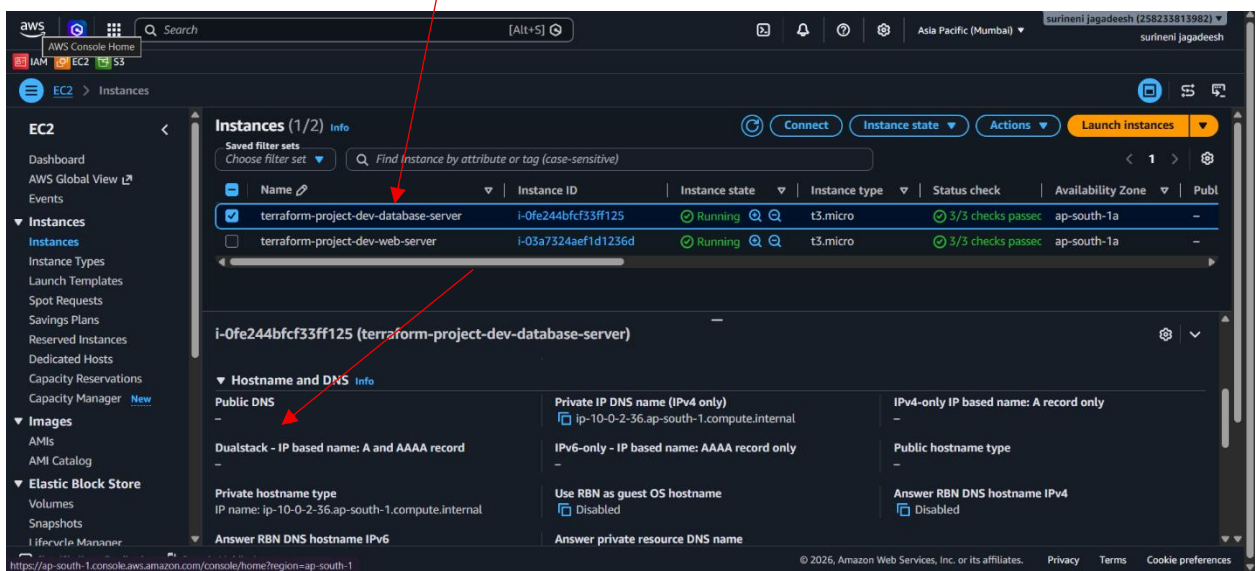
PROJECT-1

The page should show "Hello from <your-name> web server"



1 EC2 instance in the PRIVATE subnet (the database server)

No public IP



PROJECT-1

Remote backend: S3 with state locking
(use_lockfile = true)

```
aws_internet_gateway.igw: Creating...
aws_subnet.private: Creating...
aws_route_table.public: Creating...
aws_subnet.public: Creating...
aws_security_group.web_sg: Creating...
aws_internet_gateway.igw: Creation complete after 0s [id=igw-027ab774ca4129331]
aws_route_table.public: Creation complete after 0s [id=rtb-0fa41f10e35634888]
aws_subnet.private: Creation complete after 0s [id=subnet-0e29bc1324e687077]
aws_route.internet_access: Creating...
aws_route.internet_access: Creation complete after 1s [id=r-rtb-0fa41f10e356348881080289494]
aws_security_group.web_sg: Creation complete after 2s [id=sg-0dd6be22f700f7ec4]
aws_security_group.db_sg: Creating...
aws_security_group.db_sg: Creation complete after 2s [id=sg-08d0a5e1c308bca0d]
aws_instance.database_server: Creating...
```

Lock Info:

ID:	39a173f9-f4c1-b66a-0c30-55456553f470
Path:	jagadeesh-terraform-state-project-1/project-1/terraform.tfstate
Operation:	OperationTypeApply
Who:	SURINENI\jagad@SURINENI
Version:	1.15.7
Created:	2026-07-13 09:06:27.0853954 +0000 UTC
Info:	

Terraform acquires a state lock to protect the state from being written by multiple users at the same time. Please resolve the issue above and try again. For most commands, you can disable locking with the "-lock=false" flag, but this is not recommended.

PS C:\Users\jagad\OneDrive\Desktop\DevOps\terraform demo\project-1>

Screenshot of successful "terraform apply"

```
jagad@SURINENI MINGW64 ~/OneDrive/Desktop/DevOps/terraform demo/project-1
$ terraform apply
aws_route.internet_access: Creating...
aws_route.internet_access: Creation complete after 1s [id=r-rtb-0ea2014d78d2d25291080289494]
aws_security_group.web_sg: Creation complete after 3s [id=sg-029ff665019c1c554]
aws_security_group.db_sg: Creating...
aws_security_group.db_sg: Creation complete after 3s [id=sg-0e038f8a920ff3e76]
aws_instance.database_server: Creating...
aws_subnet.public: Still creating... [00m10s elapsed]
aws_subnet.public: Creation complete after 12s [id=subnet-04197b35211e5e814]
aws_route_table_association.public: Creating...
aws_instance.web_server: Creating...
aws_route_table_association.public: Creation complete after 0s [id=rtbassoc-0249ce09d9fd217511]
aws_instance.database_server: Still creating... [00m10s elapsed]
aws_instance.database_server: Creation complete after 13s [id=i-03113fece46f6cb04]
aws_instance.web_server: Still creating... [00m10s elapsed]
aws_instance.web_server: Creation complete after 12s [id=i-0612d4b6bf74034ff]

Apply complete! Resources: 11 added, 0 changed, 0 destroyed.
```

PROJECT-1

OUTPUTS:

- * Vpc_id
- * Web_public_ip
- * Web_public_url (http://<public_ip>)
- * Db_private_ip

```
jagad@SURINEMI MINGW64 ~/OneDrive/Desktop/DevOps/terraform demo/project-1
$ terraform apply

aws_instance.web_server: Still creating... [00m10s elapsed]
aws_instance.web_server: Creation complete after 13s [id=i-0194cc133984098d4]

Apply complete! Resources: 11 added, 0 changed, 0 destroyed.

Outputs:

db_private_ip = "10.0.2.148"
vpc_id = "vpc-05778b2fa0f7b97da"
web_public_ip = "43.205.235.78"
web_public_url = "http://43.205.235.78"
```

Screenshot of "terraform destroy" completed

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
jagad@SURINEMI MINGW64 ~/OneDrive/Desktop/DevOps/terraform demo/project-1
$ terraform destroy

aws_subnet.public: Destroying... [id=subnet-04197b35211e5e814]
aws_subnet.public: Destruction complete after 1s
aws_instance.database_server: Still destroying... [id=i-03113fece46f6cb04, 01m00s elapsed]
aws_instance.database_server: Still destroying... [id=i-03113fece46f6cb04, 01m10s elapsed]
aws_instance.database_server: Destruction complete after 1m11s
aws_subnet.private: Destroying... [id=subnet-08361622a8617cb03]
aws_security_group.db_sg: Destroying... [id=sg-0e038f8a920ff3e76]
aws_subnet.private: Destruction complete after 0s
aws_security_group.db_sg: Destruction complete after 1s
aws_security_group.web_sg: Destroying... [id=sg-029ff665019c1c554]
aws_security_group.web_sg: Destruction complete after 0s
aws_vpc.main: Destroying... [id=vpc-087a52f1820f43319]
aws_vpc.main: Destruction complete after 1s

Destroy complete! Resources: 11 destroyed.
```

```
jagad@SURINEMI MINGW64 ~/OneDrive/Desktop/DevOps/terraform demo/project-1
$
```